

Hanning Liu

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EDUCATION

Tsinghua University

Ph.D. student

Shenzhen International Graduate School | Institute of Future Human Habitats

Advisor: Peter John Russell

Sep. 2025 - present

Shenzhen, China

Tongji University

Master of Architecture (M.Arch)

Thesis: "Research on Mixed Reality-Assisted Design and Assembly towards Discrete Free-form Components"

Advisor: Philip F. YUAN

GPA: 4.81/5 Ranking: 1st

Sep. 2022 - Jun. 2025

Shanghai, China

Xi'an University of Architecture and Technology

Bachelor of Architecture (B.Arch)

Overall GPA: 3.93/5 Ranking: 3/192

Sep. 2017 - Jul. 2022

Xi'an, China

PUBLICATIONS

1. **H Liu**, S Wang, X Xie, PF Yuan*, "AR-Assisted Workflow in Free-Form 3D-Printed Panel Assembly: A Fusion of Virtual and Physical Elements", *CAADRIA 2025, Volume 2*, 387-396, **2025**.
2. **H Liu**, C Yan, X Xie, T Zhang, R Yang, H Wu, Y Zhang, PF Yuan*, "Bending Form in Extended Reality: A Gesture-Based Workflow of Chair Design and Fabrication", *ACADIA 2024*.
3. **H Liu**, X Xie, Y Li, X Gao, H Wu, Y Zhang, PF Yuan*, "Leveraging Motion Capture System for High Accuracy AR-Assisted Assembly", *Proceedings of CDRF*, **2024. (BEST PAPER AWARD)**
4. H Chai, L Orozco, F Kannenberg, L Siriwardena, T Schwinn, **H Liu**, A Menges*, PF Yuan*, "Agent-Based Principal Strips Modeling for Freeform Surfaces in Architecture", *Nexus Network Journal*, 26: 369–396, **2024**.
5. PF Yuan*, **H Liu**, "Spatial Implementation of Behavioral Performance: Tourist Center of Shanghai Xuhui West Coast", *Chinese and Overseas Architecture*, 01: 8-13, **2024**.
6. J Wang, **H Liu**, L Qian, H Wu, X Xie, M Yuan, PF Yuan*, "Prestressed 3D Printed Reinforced Concrete Composite Structure Design and Construction: A Case Study of Experimental Bridge Building", In *2023 Computational Design Symposium, Architectural Society of China*, **2023**. (Outstanding Paper Award)
7. **H Liu**, H Wu, X Xie, M Yuan, PF Yuan*, "Prompt Writing Approach in GAI Tools Aided Architectural Design: Taking Urban Camp Center Design as an Example", In *2023 National Symposium on Architectural Digital Technology Teaching and Research*, 433-436, **2023**.

PRESENTATIONS

1. "Bending Form in Extended Reality: A Gesture-Based Workflow of Chair Design and Fabrication", 2024 Association for Computer Aided Design in Architecture Conference (ACADIA), Banff, Alberta, Canada, **Nov 16th, 2024**.
2. "Leveraging Motion Capture System for High Accuracy AR-Assisted Assembly", the 6th International Conference on Computational Design and Robotic Fabrication (CDRF), Shanghai, China, **Jul 7th, 2024**.
3. "Prompt Writing Approach in GAI Tools Aided Architectural Design: Taking Urban Camp Center Design as an Example", 2023 National Architectural Academy Department of Architectural Digital Technology Teaching and Research Academic Symposium, Xiang Tan City, China, **Oct 14th, 2023**.

SOFTWARE COPYRIGHT

- China Software Copyright, "Fiducial marker-based AR-assisted assembly plugin for Grasshopper", 2024

RESEARCH EXPERIENCE

Residential Interior Multimodal Knowledge Bases

2026 - present

Current research (in progress)

- Developing multimodal workflows that link floor plans, interior images, text, and 3D assets.
- Representing room-zone-furniture and surface-material relations with source evidence.
- Prototyping agent-assisted extraction and human review; studying reuse of reviewed feedback.

Tongji University

Feb. 2024 - Jul. 2024

Research Assistant (with Prof. Chao Yan)

Shanghai, China

Humanizing Mixed Reality: Spatial Behavior Computation

- Designed adjustable mixed reality environments with HoloLens 2 and Quest 3 for spatial behavior experiments.
- Collected gaze, head motion, body posture, and multi-person movement data during mixed reality experiences.
- Analyzed environment-behavior relationships and visualized findings in virtual environments.

Tongji University

Sep. 2023 - Mar. 2024

Research Assistant (with Prof. Philip F. Yuan)

Shanghai, China

Pose Tracking for AR-Assisted Assembly

- Developed a Grasshopper plugin for monocular pose estimation using ArUco markers and AprilTags.
- Implemented real-time SLAM tracking with ZED 2i and RealSense D455 depth cameras.
- Integrated motion capture, custom hardware, and Unity software for accurate AR-guided assembly.

TEACHING EXPERIENCE

Guest Lecturer, Shenzhen University

Oct. 8, 2026

School of Architecture and Urban Planning

- Course: Introduction to Artificial Intelligence.
- Guest lecture: Spatial Intelligence in Residential Interior Design.
- Audience: second-year undergraduate architecture students.

Teaching Assistant for "Humanizing MR: Spatial Behavior Computation" Workshop, DigitalFUTURES Summer 2024

- Introduced cybernetics and Python simulations of human behavior and feedback rules.
- Supported human behavior data collection using HoloLens and depth cameras.

Teaching Assistant for "AI-Driven Performance-Based Tectonics" Class, Tongji University

Spring 2024

- Undergraduate and M.Arch. course: AI-assisted design with Stable Diffusion.
- Taught curved-crease folding and AR-assisted steel tube bending.

Teaching Assistant for "Computational Design" Class, Shanghai University

Fall 2023

- Undergraduate course: Rhino/Grasshopper modeling and introductory C# programming.
- Supported animation workflows with V-Ray for Grasshopper.

Teaching Assistant for ZAHA "Tectonism" Workshop, DigitalFUTURES

Summer 2023

- Supported curved-crease folding and robotic hot-wire cutting experiments.

OTHER PROFESSIONAL EXPERIENCE

Editor, DigitalFUTURES Social Media

Sep. 2022 - present

- Wrote architectural research and academic news posts, including introductions to AR research.
- Prepared linear algebra tutorials.

Webinar Host, DigitalFUTURES Webinar Series: Augmented Reality and Digital Fabrication

Nov 4, 2023

- Created promotional materials and coordinated speakers and participant registration.
- Managed scheduling, communications, and technical setup.

- Designed an exhibition building and prepared presentation materials.
- Built a physical scale model for a hotel project.
- Explored parametric brick wall design.

- Designed campus landscape spaces around a teaching building.
- Produced campus renderings.
- Designed an entrance pavilion for a residential development.

- Surveyed historic buildings using laser measurements and drone imagery.
- Produced measured plans, elevations, and sections from survey data.

- Surveyed elevations and building coordinates using leveling instruments and a total station.
- Produced 3D models and 2D drawings from survey data.

HONORS AND AWARDS

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| • Three-Good Student Award (2025-2026) , Tsinghua University | 2026 |
| • Outstanding Master's Thesis (University-Level) , Tongji University | 2025 |
| • ACADIA Conference Student Scholarship Winner , ACADIA Committee | 2024 |
| • Distinguished Design Outcome , College of Architecture and Urban Planning, Tongji University | 2023 |
| • Distinguished Design Outcome , Xi'an University of Architecture and Technology | 2021 |
| • National Encouragement Scholarship , Xi'an University of Architecture and Technology | 2019, 2020 |
| • The Third Prize , UIA-CBC International Colleges and Universities Competitive Construction Workshop | 2019 |
| • First Class Student Scholarship , Xi'an University of Architecture and Technology | 2018 |
| • The Third Prize , Harbin Institute of Technology "Ice and Snow Construction Workshop" Competition | 2018 |
| • Freshman Scholarship , Xi'an University of Architecture and Technology | 2017 |

RESEARCH INTERESTS

- Multimodal spatial understanding of residential interiors from floor plans, images, text, and 3D representations.
- Structured knowledge of room functions, furniture layouts, and surface materials for computational design.
- Human-AI collaboration, agent-assisted workflows, and evidence-aware review for spatial data construction.
- Extended reality interfaces connecting embodied design, spatial behavior, and digital fabrication.

SKILLS

Languages

- Chinese, Native (Mandarin Grade 2A)
- English, TOEFL iBT 101 (R: 29 L: 27 S: 21 W: 24)

Professional Skills

- Programming and tools: Python, C#, HTML, Git, Docker, Unity, Visual Studio.
- Spatial data: multimodal annotation, floor plan analysis, scene graphs, and 3D asset processing.
- Design: Rhino, Grasshopper, Blender, AutoCAD, Adobe Creative Cloud, V-Ray, D5.
- Fabrication: XR-assisted assembly, robotics, curved-crease folding, tube bending, hot-wire cutting, 3D printing.